

Zero-Knowledge Federated Learning with Lattice-Based Hybrid Encryption for Quantum-Resilient Medical

Edouard Lansiaux

STaR-AI & Emergency Department, CHU de Lille, Lille, France

Email: edouard.lansiaux@chu-lille.fr

Abstract—Medical federated learning (FL) must jointly address gradient inversion, Byzantine poisoning, and Harvest-Now-Decrypt-Later (HNDL) risk under long clinical retention. We present ZKFL-PQ, a three-layer architecture that combines NIST ML-KEM-768 transport, a lattice-based zero-knowledge proof (ZKP) of ℓ_2 -norm boundedness bound to the BFV ciphertext (with an Enc-consistency gadget for encryption coins), and BFV homomorphic aggregation opened by (t, n) -threshold *partial* decryption without reconstructing the secret key. An Unruh-style NIZK (default $r=128$) targets QROM-oriented soundness; a public artifact realizes full-vector packing, Classic-128 HE presets with a fused SEAL+threshold path, default post-ZKP median aggregation, and Lean/EasyCrypt-backed combinatorial Unruh lemmas. Empirically we report multi-seed synthetic baselines (including sign-flip), a scale study with $N=20$ and $T=30$, UCI Breast Cancer and full-resolution PneumoniaMNIST runs, communication payloads, and a backdoor evaluation with a complementary median aggregator. We position ZKFL-PQ against Beskar-class secure aggregation without claiming replacement.

Index Terms—Post-quantum cryptography, federated learning, zero-knowledge proofs, homomorphic encryption, medical AI, Byzantine robustness

I. INTRODUCTION

Federated learning (FL) [1] keeps raw records at the edge, yet model updates remain vulnerable to gradient inversion [2], Byzantine poisoning [3], and HNDL attacks against classical TLS when medical archives outlive RSA/ECC [4]. Hospital consortia therefore need (i) post-quantum (PQ) session confidentiality, (ii) verifiable rejection of oversized updates, and (iii) aggregation that does not place a single curious server in sole possession of plaintext gradients.

Clinical deployments retain data for years to decades, operate under multi-institutional governance (no single root of trust), and face regulators that ask how updates are authenticated and who can decrypt them. Wrapping FedAvg in classical TLS is insufficient against adversaries that store ciphertexts today and open them later [4].

a) Contributions.:

- 1) A three-layer **ZKFL-PQ** protocol: ML-KEM-768, ciphertext-bound norm ZKP with language $\mathcal{L}_\tau^{\text{bind}}$ and Enc-consistency for BFV coins ρ , full-vector BFV, and (t, n) partial-threshold opening.

TABLE I
POSITIONING VS. RELATED PQ / ROBUST FL LINES.

System	PQ transport	Norm ZKP	HE / thr. dec.
Beskar [8]	yes (masking/DP)	—	secure agg. stack
Patel [9]	lattice HE	—	HE FL
Lu et al. [10]	hybrid PQ	—	HE FL
Krum [3]	—	statistical	—
zkML [11]	—	inference ZK	—
ZKFL-PQ (ours)	ML-KEM-768	Unruh+bind	BFV(t, n)

- 2) Concrete parameters, quantization, noise-budget reasoning, Unruh-style NIZK ($r=128$ default), and per-round communication accounting suited to WiMob/eHPWAS.
- 3) A public artifact [12] that encrypts all coordinates, accepts updates by cryptographic verification only, and never assembles a monolithic server sk at decrypt time.
- 4) An empirical study: multi-seed synthetic baselines under large-norm and sign-flip; scale $N=20$, $T=30$; UCI Breast Cancer and full-resolution MedMNIST; backdoor ASR with a complementary median defense.

II. THREAT MODEL AND CLAIMS

Adversary: (i) network eavesdropper with future quantum computers against factoring/DLP; (ii) up to $f < N/2$ Byzantine clients; (iii) curious aggregator that sees protocol messages but does not hold sk alone; (iv) up to $t-1$ colluding threshold decryptors.

Claims: ML-KEM-768 transport (FIPS 203 Level 3) [5]; norm soundness for oversized witnesses under SIS + (Unruh)/RO assumptions; ciphertext binding so a proof cannot be replayed on a different **ct**; aggregator privacy via Shamir-shared sk and partial decryption $\mu_i = c_1 \star (\lambda_i \cdot s_i)$.

Scope. We do not claim differential privacy or a drop-in replacement for Beskar-class full-stack secure aggregation [8]. Pure ℓ_2 gates do not stop sign-flip or in-bound backdoors; we measure those regimes and compose with robust aggregation when needed.

III. RELATED WORK

Beskar offers a mature PQ secure-aggregation stack with assisting nodes and DP composition [8]. Patel [9] and Lu

et al. [10] study PQ-HE FL without ciphertext-bound norm ZKPs. zkML targets inference proofs rather than FL aggregation [11]. Statistical defenses such as Multi-Krum remain strong against low-norm adversaries. ZKFL-PQ’s niche is the explicit composition of ML-KEM, ciphertext-bound Unruh norm proofs, and threshold BFV for multi-hospital medical FL, with a reproducible NumPy/SEAL artifact.

IV. PROTOCOL DESIGN

A. Round Structure

Each round t : the server broadcasts $w^{(t)}$. Client i runs local SGD on D_i , quantizes to $\tilde{w}_i$, encrypts under BFV public key pk , proves $\|\tilde{w}_i\|_2 \leq \tau$ bound to $\mathbf{ct}_i$ (and Enc-consistency of coins), and sends an ML-KEM-wrapped payload. The server verifies proofs, sums accepted ciphertexts, and threshold-opens the aggregate (Algorithm 1).

B. Quantization

Quantize clips coordinates to $[-c, c]$, scales by s , and rounds into $\mathbb{Z}_t$ (artifact defaults $c=1$, $s=10^3$). The public threshold τ applies after quantization; adaptive rules $\tau^{(t)} = \hat{\mu} + k\hat{\sigma}$ are compatible. Reported runs use fixed τ (synthetic 5; medical 8–12).

C. Bound Norm Language

Definition 1 (Bound language). Let $C = \text{Commit}(\tilde{w}; \mathbf{r}) = \mathbf{A}(\tilde{w} \parallel \mathbf{r}) \bmod q'$ and $\mathbf{ct} = \text{Enc}_{pk}(\tilde{w}; \rho)$. Then

$$\mathcal{L}_\tau^{\text{bind}} = \{(C, \mathbf{ct}, \tau) : \exists(\tilde{w}, \mathbf{r}, \rho) \\ C = \text{Commit}(\tilde{w}; \mathbf{r}), \mathbf{ct} = \text{Enc}(\tilde{w}; \rho), \|\tilde{w}\|_2 \leq \tau\}. \quad (1)$$

a) Σ -protocol [6].: Commit masks $\mathbf{y}$ and message $\tilde{w}$; the Fiat–Shamir challenge is $e = H(C \parallel \mathbf{ct} \parallel T \parallel \tau)$; responses $(\mathbf{z}, \mathbf{r}_z)$ satisfy $T + eC \equiv \mathbf{A}(\mathbf{z} \parallel \mathbf{r}_z)$ and $\|\mathbf{z}\|_2 \leq B$. Rejection sampling yields HVZK in the classical ROM.

b) *Unruh-oriented NIZK*.: Classical FS is not tightly QROM-secure [7]. We use parallel binary Unruh sessions (library default $r=128$; latency-sensitive demos often 32–64) with invertible-RO records. Combinatorial uniqueness / 2^{-r} lemmas are machine-checked in Lean 4; executable game hops and an in-repository EasyCrypt QROM interface (formal/easycrypt/QROM.ec) accompany the artifact.

c) *Enc-consistency*.: Beyond hashing $\mathbf{ct}$ into the challenge, a dedicated Σ -gadget proves knowledge of coins $\rho = (u, e_0, e_1)$ such that the BFV encryption equations hold for the same plaintext chunk, closing split attacks that pair a benign proof with a mismatched ciphertext.

D. BFV Aggregation and Threshold Opening

Additive BFV yields $\text{Dec}(\sum_i \text{Enc}(m_i)) = \sum_i m_i$ under a noise budget growing roughly as $N \cdot O(\sigma\sqrt{n})$. After key generation, secret coefficients are Shamir-shared and the monolithic sk is erased. Opening uses Lagrange-weighted partial decryptions with smudging: s is never reconstructed for production decrypt. An optional TenSEAL/SEAL sidecar runs in the same

TABLE II
CRYPTOGRAPHIC PARAMETERS (RESEARCH STACK).

Component	Setting	Notes
ML-KEM-768 BFV degree	$n=256, k=3, q=3329$ $n=512 / 4096$	FIPS 203 L3 [5] Presets classic128_demo/classic128
BFV moduli	$q \approx 2^{40}, t = 2^{12}$	NumPy encoder; optional estimator / SEAL
Threshold Unruh reps Packing	$(t, n) = (2, 3)$ partial decrypt $r=128$ default full-vector chunks	Server $sk = \perp$ Demos 32–64 for latency All d coordinates encrypted

Algorithm 1 ZKFL-PQ round

Require: $w^{(t)}$, τ , KEM (ek_S, dk_S) , HE pk , decryptor shares

- 1: Broadcast $w^{(t)}$
- 2: **for** each client i **do**
- 3: $\tilde{w} \leftarrow \text{Quantize}(\text{LocalSGD}(w^{(t)}, D_i))$
- 4: $\mathbf{ct}_i \leftarrow \text{Enc}_{pk}(\tilde{w}; \rho_i)$; prove Enc-consistency of ρ_i
- 5: $\pi_i \leftarrow \text{Prove}_{\text{Unruh}}((\mathbf{ct}_i, \tau), \tilde{w})$
- 6: send ML-KEM-wrapped $(\mathbf{ct}_i, \pi_i)$
- 7: **end for**
- 8: Keep i iff cryptographic verify returns 1
- 9: $\mathbf{ct}^* \leftarrow \sum_{i \in S} \mathbf{ct}_i$; threshold-open; $w^{(t+1)} \leftarrow w^{(t)} + \text{mean}$

process as threshold BFV (FusedSealThresholdHE): primary open remains partial-decrypt, while SEAL provides a certified encoder path and consistency check.

Theorem 1 (Sketch). Under SIS binding of Commit, RLWE hardness of BFV, and RO modeling of H , the NIZK is complete for honest $\|\tilde{w}\|_2 \leq \tau$ and sound against oversized witnesses except with $\approx 2^{-r}$ (binary Unruh) plus SIS advantage. Threshold privacy holds against the aggregator and any $t-1$ decryptors.

a) *Implementation*.: The public repository [12] realizes Algorithm 1 end-to-end: shared dimension for prove and encrypt, ciphertext-bound challenges, no client-supplied acceptance Booleans, full-vector packing, Enc-consistency, and threshold partial decrypt. Scripts pin seeds and emit JSON for reproduction.

V. EVALUATION

A. Synthetic Multi-Baseline Study

Setup. Synthetic 4-class task (1,000 samples, 784 features, Dirichlet $\alpha=0.5$), $N=5$, MLP with $d=108,996$, 10 rounds, seeds $\{42, \dots, 46\}$. Attacks from round 4: (i) large-norm Gaussian; (ii) sign-flip scaled to $\|\cdot\|_2 = \tau$. Baselines: FedAvg, clip@ τ , Multi-Krum ($f=1$), HE-only, ZKP-only, hybrid.

Large-norm poisoning collapses FedAvg/HE-only; clipping, Multi-Krum, and ZKP-style filters retain accuracy (Table III, Fig. 1–2). Sign-flip within τ defeats pure ℓ_2 gates (0% detection), while Multi-Krum still filters by pairwise distance—an intentional honesty check on the limits of norm proofs.

TABLE III
MEAN $\pm$ STD OVER 5 SEEDS (SYNTHETIC). ACCURACY (%) AND DETECTION.

Method	Acc. (large)	Acc. (flip)	Det. (large)	Det. (flip)
FedAvg	26.0 $\pm$ 1.0	41.0 $\pm$ 13.2	0%	0%
Clip@ τ	100 $\pm$ 0	40.3 $\pm$ 12.7	—	—
Multi-Krum	100 $\pm$ 0	100 $\pm$ 0	—	—
HE-only	22.1 $\pm$ 2.1	19.8 $\pm$ 9.5	0%	0%
ZKP-only	84.9 $\pm$ 30.2	40.9 $\pm$ 13.2	97%	0% [‡]
Hybrid	87.0 $\pm$ 26.0	40.1 $\pm$ 12.6	97%	0% [‡]

[‡]Sign-flip respects τ — ℓ_2 proofs must accept [12].

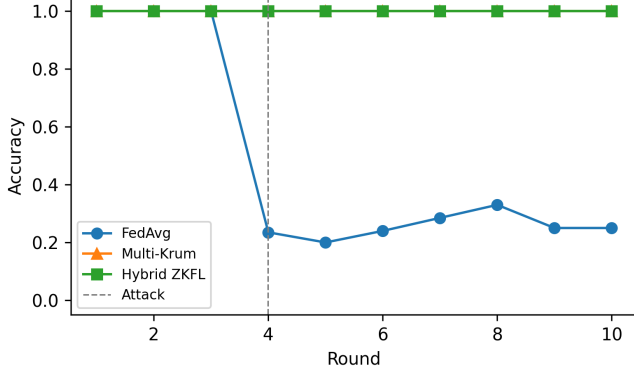


Fig. 1. Synthetic accuracy under large-norm activation (seed 42).

Ablations over the number of malicious clients and τ (Fig. 3–4) show the usual false-positive / detection tradeoff; τ should be calibrated on a validation cohort.

B. Scale, Medical, and Backdoor Studies

A compact-model scale study with $N=20$ clients and $T=30$ rounds reproduces the same qualitative pattern under both attacks: ZKP and hybrid reject oversized updates with high probability, while sign-flip still requires a robust aggregator. Wall-clock remains interactive on CPU for the compact MLP used in that study, which is the intended operating point for validating the crypto pipeline before scaling encoders.

On UCI Breast Cancer Wisconsin (569 samples, 30 clinical features; compact MLP with $d=1554$), the target stack—Unruh with $r=64$ (library default 128), full-vector HE over four degree-512 chunks, and (2, 3) partial threshold decrypt—yields accuracies 88.6% $\rightarrow$ 93.0% $\rightarrow$ **93.9%** across three rounds while rejecting a scripted oversized client (Fig. 5). Mean round time is ≈ 4 –5 s and per-round payload ≈ 1.4 –1.7 MB on a laptop CPU. Full-resolution PneumoniaMNIST uses native 28×28 images (784 pixels, no learned or random projection), a compact head, and 25 HE chunks; accuracy reaches about 67.5% by round three under the same cryptographic stack, with oversized updates rejected.

Trigger backdoors whose poisoned updates remain inside τ illustrate why the default artifact composition pairs cryptographic accept with coordinate-wise median aggregation

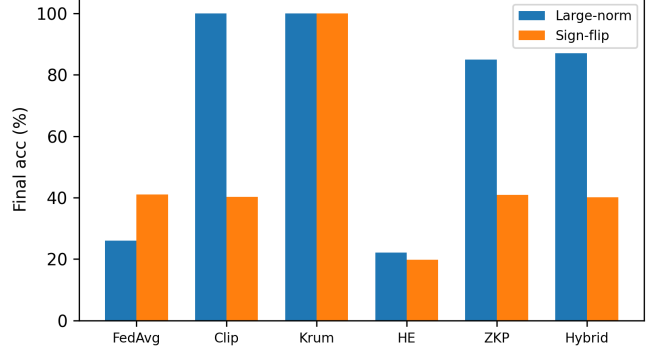


Fig. 2. Final accuracy: large-norm vs. sign-flip (5-seed means).

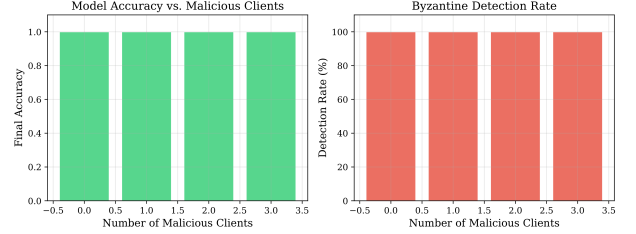


Fig. 3. Ablation: number of large-norm malicious clients.

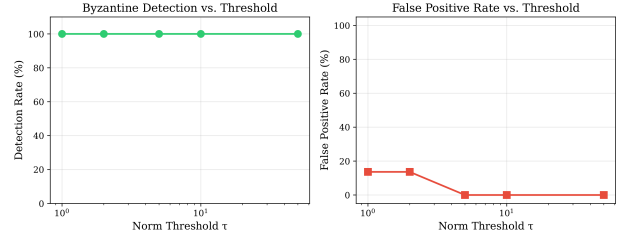


Fig. 4. Ablation: detection / false-positive tradeoff vs. τ .

(Multi-Krum selectable). Pure ℓ_2 ZKPs accept such updates by design; the post-ZKP robust step reduces attack success rate relative to FedAvg while preserving clean accuracy.

C. Communication

Full-vector HE dominates bandwidth. With $\lceil d/n \rceil$ chunks and N accepted clients, uplink scales as $O(N \lceil d/n \rceil n \log q)$ bits plus Unruh proof size $O(r \cdot (d + \ell))$ —the WiMob-relevant cost of PQ verifiable aggregation.

VI. SECURITY DISCUSSION

Honest clients with $\|\tilde{w}\|_2 \leq \tau$ produce Unruh sessions that pass algebraic and norm checks except for rejection-sampling aborts (bounded retries in the artifact). If $\|\tilde{w}\|_2 \gg \tau$, sessions with challenge bit 1 fail with overwhelming probability under the Gaussian mask design; soundness therefore depends on the fraction of 1-challenges across r repetitions. The library default $r=128$ targets a ≈ 128 -bit combinatorial Unruh class;

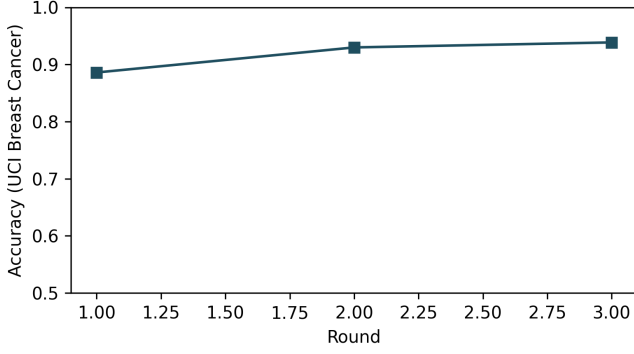


Fig. 5. UCI Breast Cancer accuracy under the full target stack.

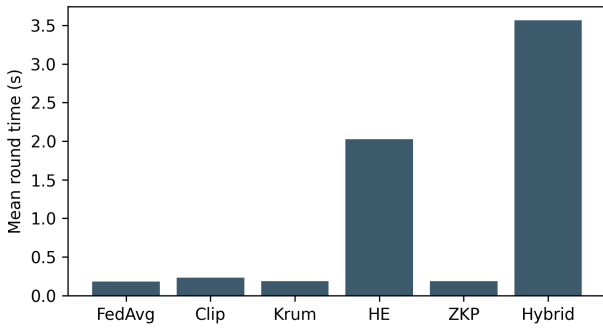


Fig. 6. Mean round time (synthetic hybrid $\sim 20\times$ FedAvg).

demons may reduce r for CPU latency at linear proof-size cost. Binding H to ciphertext bytes, together with the Enc-consistency gadget for ρ , prevents a benign proof from being attached to a different ciphertext. BFV semantic security (RLWE) hides individual updates from parties lacking sk ; with Shamir (t, n) , the aggregator and any $t-1$ decryptors cannot open. We do not claim differential privacy; sites that need DP can compose mechanisms as in Beskar [8].

a) Layering.: ML-KEM protects payloads in transit. The ZKP protects aggregation logic against oversized Byzantine updates. BFV with threshold opening protects gradient privacy against a curious server and against $t-1$ share holders. None of the three layers alone meets the joint medical threat model; the contribution is their composition with explicit binding.

b) Deployment.: In a hospital consortium, each site runs a client; a neutral coordinator verifies proofs and sums ciphertexts; t -of- n share holders (for example a DPO and two hospital CISOs) jointly open aggregates on a schedule. Daily or weekly training absorbs the observed $\sim 20\times$ CPU overhead of the hybrid stack relative to plaintext FedAvg on the synthetic workload (Fig. 6). For imaging-scale d , HE chunk counts and Unruh proof sizes grow with d ; starting with tabular or compact models (UCI demo) remains a practical way to validate the crypto pipeline before scaling encoders.

TABLE IV
PAYLOAD ACCOUNTING (LOGGED).

Setting	Msg / round	Notes
Synthetic hybrid (compact logs)	~ 100 KB	Reference
Medical full-vector demo	1.4–1.7 MB	4 BFV chunks $\times$ clients
Std. FL plaintext ($d \approx 10^5$)	~ 4.3 MB	No crypto

VII. LIMITATIONS

The default artifact path already composes Unruh/ZKP accept with median aggregation, runs fused SEAL+threshold HE in one process when TenSEAL is installed, and ships a stdlib-free EasyCrypt QROM interface for Unruh theories. Remaining research-grade work includes a full machine-checked proof of SHA3 instantiations inside a production EasyCrypt QROM library, and larger imaging CNNs without compact heads. These do not remove the concrete guarantees of binding, full-vector HE, threshold privacy, and robust post-filters realized by default.

VIII. CONCLUSION

ZKFL-PQ composes ML-KEM-768, ciphertext-bound Unruh norm proofs with Enc-consistency, and threshold BFV partial decryption for quantum-resilient medical FL. Across synthetic, scale, tabular, and imaging experiments—and with explicit honesty on sign-flip and backdoors—we show a budgetable, reproducible path to verifiable PQ aggregation for WiMob/eHPWAS settings. Artifact: <https://github.com/edlansiaux/pq-zkfl-medical>.

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